Session 10 — Non-Current Assets II: Subsequent Costs, Measurement After Recognition and Investment Property (IAS16 / IAS40)

By the end of this session you must know how to classify subsequent PP&E expenditure as capitalised or expensed, the mechanics of the cost model and the three depreciation methods, how the revaluation model works including its surplus/deficit rules, and how IAS40 treats investment property differently from owner-occupied property under IAS16.

Plan

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1. Subsequent Costs: Capitalise or Expense?

Subsequent costs (IAS16 par 13): after an asset has been installed and made ready for use, a company incurs additional costs ranging from ordinary repairs to significant additions. These can be recognised as an asset only if it is probable that future economic benefits associated with the item will flow to the enterprise and if the cost can be reliably determined.

Must know (capitalise vs. expense):

• **Capitalise:** additions that increase efficiency, capacity or useful life; improvements (substitution with a better asset) or replacements (substitution with a similar asset), when the expenditure increases the asset's future service potential; major (non-routine, e.g. legally mandated) inspections

• **Expense:** ordinary repairs and maintenance; re-arrangements and re-organisation (moving assets from one location to another) — IFRS treats the recognition of such costs as ceasing once the asset is in the location and condition intended by management

- When a component is replaced, the standard approach is to derecognise the carrying amount of the old component (and its associated depreciation, recognising any resulting loss) and then capitalise the cost of the new component — this is the practical application of the component approach introduced in Session 9.

Wall lining replacement: a company pays €80,000 to replace the wall lining of a furnace acquired seven years earlier. Before replacement the furnace's carrying amount was €320,000, of which €10,000 related to the original lining. The new lining is capitalised and the old lining's remaining carrying amount is derecognised: new carrying amount = 320,000 + 80,000 − 10,000 = €390,000.

Major aircraft inspection: a legally required inspection (at least once every three years) costs €250,000 on 1 September; the previous inspection (March, two and a half years earlier) cost €210,000. The aircraft's carrying amount before the new inspection is €1,200,000, of which €70,000 relates to the previous inspection still carried. New carrying amount = 1,200,000 + 250,000 − 70,000 = €1,380,000. The logic mirrors the wall-lining example: capitalise the new major cost, derecognise the remaining carrying amount of what it replaces.

2. Measurement After Recognition: Cost Model and Depreciation

Measurement after recognition (IAS16 par 29): an entity chooses, as an accounting policy applied to an entire class of PP&E (not asset by asset), between the cost model and the revaluation model.

Cost model (par 30): after recognition, an item of PP&E is carried at its cost less accumulated depreciation and less accumulated impairment losses.

Depreciation rules (par 43-56):

- Depreciation must be charged on a systematic basis over the asset's useful life.
- Useful life is determined considering expected usage, expected physical wear and tear, expected technical or commercial obsolescence, and legal or similar limits on use.
- The depreciable amount is cost less residual value.
- The depreciation method and useful-life estimate must be reviewed each financial year, with any change treated as a change in accounting estimate (an ordinary, not extraordinary, component of income — applied prospectively, consistent with IAS8, Session 3).
- Impairment losses are recognised per IAS36 (Session 12) by comparing the net carrying amount with the recoverable amount

(the higher of value in use and fair value less costs to sell), recognised in profit or loss.

- Land is not depreciated and must be accounted for separately from any building purchased jointly with it.

Three depreciation methods:

Must know (depreciation methods):
• **Straight-line:** a constant charge over the asset's useful life
• **Diminishing-balance:** a decreasing charge over the asset's useful life — appropriate when productivity/revenue-earning power is relatively greater in early years, or maintenance charges rise in later years
• **Units-of-production:** a charge based on the asset's expected use or output

Worked example (machine, cost €40,000, residual value €1,024, useful life 4 years, total production 9,744 units over years of 2,260 / 2,430 / 2,550 / 2,504 units):
Straight-line: depreciable amount = 40,000 − 1,024 = 38,976; annual charge = 38,976 ÷ 4 = €9,744.
Diminishing-balance (60% rate): Year 1 charge = 60% × 40,000 = 24,000 (NBV 16,000); Year 2 charge = 60% × 16,000 = 9,600 (NBV 6,400); Year 3 charge = 60% × 6,400 = 3,840 (NBV 2,560); Year 4 charge = 60% × 2,560 = 1,536 (NBV 1,024 = residual value, exactly).
Units-of-production: rate per unit = 38,976 ÷ 9,744 = €4/unit; charges = 4 × 2,260 = 9,040 (Year 1); 4 × 2,430 = 9,720 (Year 2); 4 × 2,550 = 10,200 (Year 3); 4 × 2,504 = 10,016 (Year 4).

3. The Revaluation Model

Revaluation model (par 31-36): an item of PP&E is carried at fair value at the date of revaluation, net of subsequent accumulated depreciation and any subsequent impairment losses.

Two ways to apply a revaluation are possible:

- **Option A, gross-value revaluation** — restates both the gross cost and the accumulated depreciation proportionally (typically used with a revaluation coefficient).
- **Option B, net-value revaluation** — restates only the net carrying amount directly (typically used for buildings).

Example: a plant has cost 100, accumulated depreciation 40, and fair value 90. Under Option A, gross value becomes 150 and accumulated depreciation becomes 60 (net 90, same result). Under Option B, the plant is simply restated to its net value of 90.

- The revaluation model must be applied to an entire class of PP&E — an entity cannot selectively revalue only some assets within a class.
- Revaluations must be carried out regularly, though annual revaluation is not required if there are no significant fluctuations; every three to five years is sufficient in that case.
- Assets carried at a revalued amount continue to be depreciated (only the depreciation rate/base may need adjusting), which is an important contrast with investment property held at fair value under IAS40 (Section 5).

4. Revaluation Surplus/Deficit Mechanics

This is the single most exam-relevant mechanical rule in this session:

Must know (revaluation surplus/deficit rule):
• **First revaluation, surplus:** credited to a revaluation reserve via OCI (does not touch profit or loss)
• **First revaluation, deficit:** debited to profit or loss
• **Subsequent revaluation, surplus:** credited to the revaluation reserve via OCI, *unless* it reverses a previous deficit that was charged to profit or loss — in that case, the surplus is credited to profit or loss up to the amount of the previous deficit, and only any excess goes to the revaluation reserve
• **Subsequent revaluation, deficit:** debited to profit or loss, *unless* it reverses a previous surplus sitting in the revaluation reserve — in that case, the deficit is first debited against that reserve balance, and only any excess beyond the reserve balance is debited to profit or loss

Land revaluation (Company X): land acquired for €400,000. Revalued to €450,000 the following year (first revaluation, a surplus of €50,000 → €50,000 credited to the revaluation reserve via OCI). Revalued again the next year to €375,000 (subsequent revaluation, a deficit of €75,000: €50,000 first reverses the reserve balance down to zero, and the remaining €25,000 is debited to profit or loss).

Land revaluation (Company Y): land acquired for €600,000. Revalued to €540,000 (first revaluation, a deficit of €60,000 → debited entirely to profit or loss, since there is no prior surplus to reverse). Revalued again to €620,000 (subsequent revaluation, a surplus of €80,000: €60,000 first reverses the prior P&L deficit and is credited back to profit or loss, and the remaining €20,000 is credited to the revaluation reserve via OCI).

Watch out: the extracted slide for the Company Y solution literally reads “credit to debit to revaluation reserve” for the residual €20,000 — almost certainly a copy-paste typo in the source slide. Based on the surplus/deficit rule and the parallel structure of the Company X example, the correct treatment is a *credit* to the revaluation reserve (via OCI), which is what is stated above; verify this against the instructor’s slide if in doubt.

Multi-year fair-value table exercise: an asset costing 100,000 with a 5-year useful life is fair-valued at year-end after each year’s straight-line depreciation charge is booked. Year 202X: gross 100,000, depreciation 20,000, net value 80,000, fair value 90,000 → surplus of 10,000 to the revaluation reserve. Year 202X+1: opening 90,000, depreciation 22,500 (90,000 ÷ 4 remaining years), net 67,500, fair value 59,400 → deficit of 8,100, absorbed entirely by the 10,000 reserve (reserve falls to 1,900). Year 202X+2: opening 59,400, depreciation 19,800, net 39,600, fair value 35,000 → deficit of 4,600 in the source table’s own subtotals absorbs the remaining 1,900 reserve then charges 2,700 to profit or loss (reserve at zero). Year 202X+3: opening 35,000, depreciation 17,500, net 17,500, fair value 25,000 → surplus of 7,500, of which 2,700 first reverses the prior P&L deficit (credited back to profit or loss) and 4,800 is credited to the reserve. Year 202X+4: asset fully depreciated to zero, reserve remains at 4,800.

5. IAS40: Investment Property in Brief

Objective of IAS40 (par 1): to prescribe the accounting treatment for investment property and related disclosures, since property acquired as an investment (rather than for use) is not consumed in operations and does not have a “useful life” in the IAS16 sense — IAS16 is generally inappropriate for such property, so IAS40 provides a more suitable treatment.

Investment property (par 5): land or a building (or part of a building), held by the owner or by a lessee as a right-of-use asset, to earn rentals or for capital appreciation, or both. This is distinguished from owner-occupied property, which is held for use in the production or supply of goods or services or for administrative purposes and remains within IAS16.

- On acquisition, investment property is initially measured at cost, exactly as under IAS16.
- Subsequently, the entity chooses between the cost model (depreciate and write down for lasting impairment, exactly as in Section 2) and the fair value model.

Must know (IAS40 fair value model, contrast with IAS16 revaluation):
• Fair value is remeasured **every year** (not every 3–5 years as under IAS16’s revaluation model)
• The property **ceases to be depreciated** once fair-valued (unlike PP&E under the IAS16 revaluation model, which continues to be depreciated)
• Both increases **and** decreases in fair value pass through **profit or loss** directly — there is no OCI/revaluation-reserve route, unlike IAS16
• The accounting policy must be applied consistently to the entire class (all land and buildings held as investments)

6. Common Mistakes and Exam Traps

- Applying the IAS16 revaluation surplus/deficit (OCI/reserve) mechanics to investment property under the IAS40 fair value model — IAS40 fair-value gains and losses always go through profit or loss, never through a revaluation reserve.
- Continuing to depreciate an investment property measured at fair value under IAS40 — this is the opposite of PP&E under the IAS16 revaluation model, where depreciation continues.
- Forgetting that a revaluation surplus reversing a prior P&L deficit is credited back to profit or loss (not straight to the reserve) up to the amount of that prior deficit, and only the excess goes to the reserve — and symmetrically for a deficit reversing a prior reserve surplus.
- Treating “major repairs” the same as “ordinary repairs” — major/non-routine (e.g. legally mandated periodic) inspections are capitalised, ordinary maintenance is expensed.
- Assuming a change in an asset’s depreciation method or useful-life estimate requires restating prior periods — under IAS8 (Session 3) this is a change in accounting estimate, applied prospectively only.

7. Must Know Absolutely

Must know:
• Subsequent costs capitalised if they increase future economic benefits (additions, improvements, replacements, major inspections); ordinary repairs and relocation/reorganisation costs are expensed
• Cost model: cost − accumulated depreciation − accumulated impairment
• Three depreciation methods: straight-line (constant), diminishing-balance (decreasing), units-of-production (usage-based)
• Revaluation model: fair value − subsequent depreciation/impairment, applied to an entire asset class, assets continue to be depreciated
• Surplus/deficit rule: first surplus → OCI/reserve; first deficit → P&L; subsequent moves reverse the prior one first (deficit-reversing-surplus and surplus-reversing-deficit both split between reserve and P&L at the point where the prior balance is exhausted)
• IAS40 investment property: rentals/capital appreciation purpose (vs. IAS16's own-use purpose); fair value model = annual remeasurement, no depreciation, gains/losses through P&L (not OCI)

Watch Out

Watch out: the Company Y revaluation example's slide text contains an apparent typo ("credit to debit to revaluation reserve") for the residual €20,000 after reversing the prior deficit; Section 4 above resolves it as a credit to the reserve via OCI, consistent with the surplus/deficit rule and the mirrored Company X example, but confirm this against the instructor's own slide or working.